A counterexample to the Meijer-function zero-count conjecture

Principia Math in Progress

13 September 2026

Task 1. Resolve the zero-count conjecture. Conjecture 1 of Karp and Prilepkina [1] asserts that, for $a_i \geq 0$ and $\sum_i (b_i - a_i) > 0$, the number of zeros of $G^{p,0}_{p,p}(x \mid b/a)$ in $(0,1)$ does not exceed the number of zeros of $v(t) = \sum_i (t^{a_i} - t^{b_i})$ there. We refute it with strictly positive rational parameters. The zero count of an explicit polynomial is certified by finite integer arithmetic and Descartes' rule of signs.

Theorem 1. *Let*

$$a = \left(1, \frac{1999}{1000}, \frac{3099}{1000}, \frac{3447}{1000}\right), \qquad b = \left(5, \frac{999}{1000}, \frac{2099}{1000}, \frac{2447}{1000}\right).$$

All parameters are positive and $\sum_i (b_i - a_i) = 1$. The corresponding Meijer function has exactly three distinct zeros in $(0,1)$, whereas v has exactly one, which is simple.

Proof. Put $r = (-1, 1099, 1447)/1000$. The gamma recurrence gives

$$\prod_{i=1}^{4} \frac{\Gamma(s + a_i)}{\Gamma(s + b_i)} = \frac{\prod_{j=1}^{3}(s + 1 + r_j)}{(s+1)(s+2)(s+3)(s+4)}.$$

By partial fractions, the Mellin transform on $(0,1)$ of $xF(x)/(6 \cdot 10^9)$ is this same rational function, where

$$F(x) = 8859711253x^3 - 2991012759x^2 + 132891759x - 1590253.$$

Mellin inversion, or direct evaluation of the defining rational Mellin–Barnes integrand, therefore gives

$$G^{4,0}_{4,4}(x \mid b/a) = \frac{xF(x)}{6 \cdot 10^9}, \qquad 0 < x < 1.$$

This identification involves no numerical evaluation of a special function. The following four exact values have alternating signs:

$$F(0) = -1590253, \quad F(1/40) = \frac{69823293}{64000}, \quad F(3/100) = -\frac{56199509269}{10^6}, \quad F(3/10) = \frac{8298330221}{1000}.$$

Thus the cubic has three distinct interior zeros, one in each of the consecutive intervals determined by these points. Its degree shows that there are no further zeros.

For the comparison function, put $u = t^{1/1000}$. Directly from the parameter vectors,

$$v(t) = u^{999}(1 - u^{1000})q(u), \qquad q(u) = -1 + u + u^{1001} - u^{1100} - u^{1448} + u^{2001} + u^{3001}. \tag{1}$$

The prefactor is strictly positive for $0 < u < 1$, so it remains to count the zeros of q on that interval.

Here is the exact integer certificate. Let $N = 3001$, and for integers $0 \leq A < C \leq B$ define

$$H_{A,C,B}(y) = (B(1+y))^N q\left(\frac{C + Ay}{B(1+y)}\right) \in \mathbb{Z}[y].$$

The fractional-linear substitution maps $y > 0$ bijectively onto $A/B < u < C/B$. Multiplication by its positive prefactor preserves zeros and their multiplicities. For each row of the table below, the coefficient signs are listed

in increasing degree; none is zero.

A	C	B	interval	coefficient sign string of H
0	99	100	$(0, 99/100)$	$-^{3002}$
990	995	1000	$(99/100, 199/200)$	$-^{3002}$
995	997	1000	$(199/200, 997/1000)$	$-^{3002}$
997	998	1000	$(997/1000, 499/500)$	$-^{3002}$
998	999	1000	$(499/500, 999/1000)$	$-^{3002}$
999	1000	1000	$(999/1000, 1)$	$+^{2587}-^{415}$

Here $-^m$ and $+^m$ mean m consecutive minus and plus signs, respectively. The table is reproducible with the integer recurrence below. Two independently implemented coefficient constructions agree on all $6 \cdot 3002 = 18012$ coefficients.

Descartes' rule gives no roots in the first five intervals and at most one root, counted with multiplicity, in the last. Exact integer evaluation gives $q < 0$ at all six left endpoints in the table and $q(1) = 1$. There are no roots at a subdivision point. The intermediate value theorem supplies a root in the last interval, so there is exactly one and it is simple. Equation (1) proves the claimed count for v, and $3 > 1$ disproves the conjecture. □

Reproducing the finite certificate. Write $q(u) = \sum_i c_i u^i$, where its seven nonzero coefficients are displayed in (1). A summand of H is $c_i B^{N-i}(C + Ay)^i (1 + y)^{N-i}$. If its coefficients are h_k, set

$$h_{-1} = 0, \qquad h_0 = c_i B^{N-i} C^i, \qquad \ell = iA + (N - i)C.$$

Differentiating this product gives the recurrence

$$h_{k+1} = \frac{(\ell - (A + C)k)h_k + A(N - k + 1)h_{k-1}}{C(k + 1)}, \qquad 0 \le k < N. \tag{2}$$

Sum these coefficient lists over the seven values of i. Every division in (2) is exact. The supplied `verify.py` checks this, verifies the complete sign strings and endpoint signs, and independently reconstructs H by polynomial composition and reversal. Thus the root count covers the entire real interval, including multiplicities; sampled plots and floating-point root estimates are not used in the certificate.

Scope. The example lies strictly within the nonnegative-parameter and positive-excess hypotheses. Taking all scale parameters equal to one also refutes the more general Fox-function assertion in Conjecture 3. This immediate consequence receives no additional campaign credit.

Task 2. Find smaller counterexamples. Determine the least parameter count p for which the conjecture fails, and seek examples with smaller parameter denominators.

Task 3. Find a replacement zero bound. Determine whether an alternative bound in terms of p, the parameter ordering, and the zeros of v holds for positive parameters.

Task 4. Describe counterexample regions. Identify explicit open regions of parameter space where the Meijer function has three simple interior zeros and v has one.

Task 5. Study nonunit Fox scales. Construct counterexamples with unequal positive scale parameters and analyze how the two zero counts change as those scales vary.

References

[1] D. B. Karp and E. G. Prilepkina, *Completely monotonic gamma ratio and infinitely divisible H-function of Fox*, Comput. Methods Funct. Theory **16** (2016), 135–153, doi:10.1007/s40315-015-0128-9; arXiv:1501.05388v2. Conjecture 1 is on p. 5 of the 18-page arXiv PDF and p. 140 (PDF page 8) of the inspected author-hosted journal offprint.